

# CHAEYUN KIM

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## RESEARCH INTERESTS

Visual Grounding, Generative AI (Diffusion-based Image Editing), Multi-modal Understanding (Compositional Understanding, Retrieval), Visual-Linguistic Concept on Motion/Action, Video Temporal Grounding

## EDUCATION

### Seoul National University (SNU)

Seoul, Korea

*M.S. in Data Science*

Feb. 2025 (expected)

- Relevant Coursework: Machine Learning & Deep Learning I, II, Big Data & Knowledge Management Systems I, II, Computing for Data Science I, Scalable High-Performance Computing, Data Science Project, Machine Learning for Visual Understanding, Conversational AI & Dialogue Systems

*B.S. in International Relations*

Mar. 2017 – Feb. 2022

- CGPA: 4.15/4.3, Graduated *Summa Cum Laude*
- Dean's List, Academic Scholarship for 5 semesters (2018-2020)

## EXCHANGE PROGRAMS

### University of Washington, Seattle

Seattle, WA

*Henry M. Jackson School of International Studies*

Mar. 2020 – Jun. 2020

### Vrije Universiteit Brussel

Brussels, Belgium

*Institute for European Studies*

Jul. 2019 – Aug. 2019

- Participated in SNU in the EU program; took a lecture on the European Union's international mission.

## RESEARCH EXPERIENCE

### SNU Visual Information Processing Lab

Seoul, Korea

*Graduate Researcher*

Sep. 2024 – Feb. 2025

- Research compositional understanding in vision-language tasks, especially motion and action concept decomposition; two projects in progress under preparation for submissions to ICCV 2025:
  - 1) “Unified Framework for Action-Centric Query Understanding in Visual Grounding” (Advisor: Prof. Joonseok Lee, Prof. Yohan Jo): Develop a hierarchical regularization method for motion-centric queries using verb-centric positive examples and granularity-driven negative examples.
  - 2) “Mosaic Puzzle Pre-training for Motion Understanding in Referring to Video Segmentation”: Design a dual framework that integrates mined video mosaics and temporal dynamics prediction to improve motion comprehension.

*Graduate Research Assistant*

Jan. 2023 – Aug. 2024

- Designed a video frame interpolation model with Fourier Transform loss to preserve high-frequency details in low-FPS sampled frames, using multi-resolution inter-frame deformable attention to enhance the representation of diverse object features.
- Developed negative-mined mosaic augmentation for referring image segmentation, improving model comprehension of visual-linguistic cues in ambiguous scenarios; co-first author of an ECCV 2024 paper.

## PUBLICATIONS

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[Journal]

Ha, S.S. \* **Kim, C.Y.**, \* Kim, D.H., \* Lee, J.H., Lee, S.H., Lee, J.S. “Finding NeMo: Negative-mined Mosaic Augmentation for Referring Image Segmentation,” *ECCV, 2024* (\*denotes equal contributions)

[Books]

Kim, S.B., **Kim, C.Y.**, et al, “The 4th Industrial Revolution and U.S.-China Power Politics: A Perspective from Information Politics,” Social Criticism Academy, May 2020 (authored subsection of Chapter 8), [Link](#)

Kim, U.Y., **Kim, C.Y.**, et al, “2020 Civic Politics Yearbook: Theories and Practices of Community-based Education,” Pureungil, Jan. 2021 (authored Chapter 3), [Link](#)

## PROJECT EXPERIENCE

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“Resolving Multimodal Hallucinations through Semantics-Guided Augmentation”

*Team Leader*

Sep. 2023 – Dec. 2023

- Defined multimodal hallucination as inconsistencies in output when semantically identical but visually different texts are provided in a Visual Grounding task.
- Identified ambiguous or overly simplistic linguistic annotations in datasets as the primary cause, focusing on improving text modeling to enhance reasoning capabilities.

“ECOLENS: Real-time Object Recognition for Recycling Guidance”

*Machine Learning Engineer & Back-end Developer*

Sep. 2023 – Dec. 2023

- Developed a system providing user-customized text and speech recycling instructions for objects based on photos and information about available bins.
- Constructed the application backend using the Flask framework; implemented a MongoDB database for efficient data management.

“Data-Effective Semantic Segmentation on 4D Point Cloud Videos”

*Team Member*

Mar. 2023 – Jun. 2023

- Designed a data-loading pipeline optimized for processing large-scale datasets.
- Proposed a framework combining active learning and contrastive learning to efficiently utilize feature representations in irregular and unordered point clouds.

“GSDS PleaseGraduate: Graduation Requirement Checker and Course Recommender”

*Team Leader*

Oct. 2022 – Jan. 2023

- Spearheaded the development of a customized graduation requirement checking system for the Graduate School of Data Science.
- Designed user interface and led front-end development using HTML, CSS, jQuery, and Django; managed the data preprocessing pipeline.

## TEACHING ASSISTANTSHIPS

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“Core Computing: Thinking with Computers – Data Structure and Algorithms,” SNU

Fall 2024

“Principles and Applications of Data Science,” SNU

Fall 2023

“Machine Learning and Deep Learning I,” SNU

Fall 2023

“Basic Computing: First Adventures in Computing,” SNU (tutor)

Fall 2022

## SCHOLARSHIPS & AWARDS

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### Graduate

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|-----------------------------------------------------------------------------|-----------|
| <i>Full-tuition Academic Scholarship</i> , Seoul National University        | Sep. 2023 |
| <i>Data Science Graduate School Scholarship</i> , Seoul National University | May 2023  |
| <i>4<sup>th</sup> Stage BK21 Scholarship</i> , Seoul National University    | Mar. 2023 |

### Undergraduate

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|---------------------------------------------------------------------------------------|-----------------------|
| <i>Study Abroad Scholarship</i> , SNU Office of International Affairs                 | Sep. 2020             |
| <i>Dean's List &amp; Academic Scholarship</i> , Seoul National University             | Sep. 2018 – Dec. 2020 |
| <i>Global Leadership Award</i> , Seoul National University                            | Feb. 2022             |
| <i>1st Place</i> , Division of Policy Strategy, Future Aerospace Conference           | Oct. 2021             |
| <i>1st Place</i> , Undergraduate Social Science Research Grant, SNU                   | Feb. 2020             |
| <i>3rd Place</i> , KOICA 16th International Development Cooperation Paper Competition | May 2019              |

## LEADERSHIP

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| <i>Vice President</i> , SNU Graduate School of Data Science Student Association                                                                                                                                                                                                                             | Aug. 2023 – Feb. 2024 |
| <ul style="list-style-type: none"><li>Served as the cohort's deputy leader and an active member, representing the Data Science department.</li><li>Organized various on-campus events, bi-weekly student seminars, and community-building activities in collaboration with school administrators.</li></ul> |                       |

## ACADEMIC ACTIVITIES

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| <b>SNU-HIT International Joint Seminar</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sep. 2017 – Oct. 2017 / Sep. 2019 – Oct. 2019 |
| <ul style="list-style-type: none"><li>Participated in the international exchange seminar between Seoul National University Department of Political Science &amp; International Relations and Hitotsubashi University Faculty of Law<ul style="list-style-type: none"><li>2017: Presented and led discussions on the "Korea-Japan Security Cooperation Plan" and "General Security of Military Information Agreement (GSOMIA) and Japan's Constitutional Amendment."</li><li>2019: Hosted seminars in the History Department; facilitated discussions on the Supreme Court's ruling on forced labor, individual claim issues, and strategies to improve the Korea-Japan relations."</li></ul></li></ul> |                                               |
| <b>12<sup>th</sup> Asian Future Political Leaders' Association (AFPLA)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | May 2018 – Aug. 2018                          |
| <ul style="list-style-type: none"><li>Served as Politics Division Session Leader; presented and led discussions on "Definition, Importance, and the Evolution of 'Citizen' and 'Citizen Culture' in Past, Present, and Future Contexts."</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                               |

## PROFICIENCY IN SKILLS

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**Programming:** Python, R, SQL, C/C++

**Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Sckit-learn, Seaborn, Flask

**Languages:** Korean (native), English (full-professional fluency), Mandarin Chinese, Spanish (intermediate)